



Full length article



Automated segmentation and analysis of microscopy images of laser powder bed fusion melt tracks

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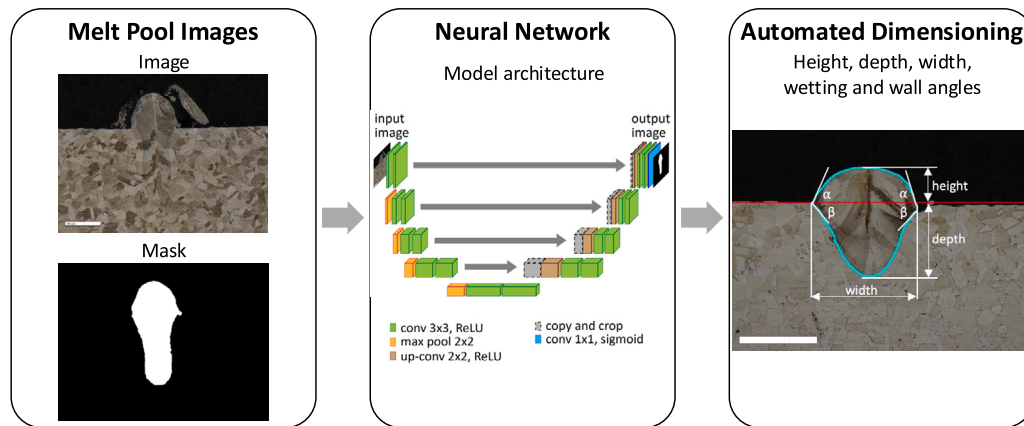
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GRAPHICAL ABSTRACT



HIGHLIGHTS

- Trained a model to segment optical micrographs of LPBF melt track cross-sections.
- The model has an accuracy greater than 99% and F1 score greater than 90%.
- Associated software converts the segmented images into quantitative metrics.
- The model and software are available as an installable Python package.
- The model can be used for transfer learning to apply it to other AM techniques.

ARTICLE INFO

Keywords:

Laser powder bed fusion
Machine learning

ABSTRACT

With the increasing adoption of metal additive manufacturing (AM), researchers and practitioners are turning to data-driven approaches to optimise printing conditions. Cross-sectional images of melt tracks provide valuable information for tuning process parameters, developing parameter scaling data, and identifying defects. Here we present an image segmentation neural network that automatically identifies and measures melt track

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dimensions from a cross-section image. We use a U-Net architecture to train on a data set of 62 pre-labelled images obtained from different labs, machines, and materials coupled with image augmentation. After tuning the neural network hyperparameters, the learned model has a pixelwise accuracy of over 99% and an F1 score over 90%. The neural network exhibits robustness when tested on images captured by various users, printed on different machines, and acquired using different microscopes. A post-processing module extracts the height and width of the melt pool, and the wetting angles. We discuss opportunities to improve model performance and avenues for transfer learning, such as extension to other AM processes such as directed energy deposition.

1. Introduction

Additive manufacturing (AM) by laser powder bed fusion (LPBF) can create near net shape components with intricate geometries, without specialised tooling, and finds commercial applications in aviation components, medical implants, and bespoke mechanical systems, among others [1–4]. One of the most promising avenues in LPBF is the development and application of new, tailored materials, including advanced designs such as multi-material gradient structures [5–7].

LPBF practitioners that want to optimise processing strategies (e.g., to improve production rate or the quality of parts) or utilise new materials have to develop suitable processing parameters. This is often pursued via iterative trial and error experimentation. Laser power, scan speed, laser spot size, hatch spacing, and powder layer thickness are among the parameters that have to be optimised for a successful build. The initial parameter space is often very large, requiring a large number of experiments and analysis.

While physics-based computational models are not yet sufficiently developed and available at scale, data-driven optimisation methods are a promising avenue to support parameter development [8–10]. To generate a sufficient amount of data for this purpose, single layer and single track experiments are significantly more efficient to perform than full builds when doing parameter development studies. Exemplary research directions informed by single-layer and single-track studies are powder spreading [11–16], melt pool dynamics and scaling [12,17–31], new material development [32], microstructure evolution [28,33], and formation of porous materials in LPBF [34].

To interpret results from single-track experiments, manual analysis of optical microscope images of the melt track cross-sections is typically performed. These images provide quantitative information that can be used to understand the manufacturing regime, such as the height and width of the melt pool, as well as wetting angles, which makes the analysis a critical step in generating research results. Manual image analysis is time consuming and can be prone to human error. Moreover, data-driven approaches to interpret LPBF, such as through scaling relationships among dimensionless process parameters, requires a large number of cross-sectional images [35–41]. Accordingly, in recent years, machine learning and deep learning techniques have been employed to simplify the task of analysing different types of images of metal AM techniques [42–50]. The Segment Anything Model (SAM) by Meta AI [50] is a notable example. As its name suggests, it is designed for general-purpose image segmentation across a wide variety of domains. However, in the context of large-scale workflows, several limitations become apparent: (i) the model's substantial size makes it computationally demanding and slow; (ii) it still requires user input to determine the appropriate segmentation level for each image or region; and (iii) effective use of SAM typically demands some familiarity with programming and machine learning concepts. Many of the techniques above have been developed specifically for LPBF, but most of them do not tackle single-track cross-sectional images [42,46,47]. While Fody et al. have demonstrated a model for single-track images [51], one capable of handling images from diverse materials systems remains elusive.

In this work, we demonstrate the application of U-Net to automated segmentation of optical microscopy images of cross-sections of melt tracks from LPBF experiments. Our strategy used images of single melt

track obtained from two research labs using different machines, sample preparation approaches, and microscopes to build a versatile training data set. We developed software to aid with the generation of training data based on a combination of existing unsupervised techniques. We train a U-Net-based neural network, with training images generated with the help of our software, to segment cross-sectional images of LPBF melt tracks into the region of the melt track and the surrounding region, and show that it can automatically and rapidly quantify the height, depth, width, wetting angles, and wall angles of a melt pool. The model trained on only single track images is also able to segment some multi-track images with reasonable accuracy indicating the possibility of few-shot or transfer learning. This work presents an automated technique for the analysis of optical microscopy images of cross-sections of LPBF melt tracks, towards fostering closed loop optimisation of LPBF. The trained model is available for general use, and provided with an installable software for ease of use [52].

2. Methods

2.1. Experiments and imaging

The employed neural network architecture was trained on a curated data set of cross-sectional images of single line melting experiments. Images from three sources were used: bare plate melting experiments performed using a bespoke miniature LPBF testbed [53]; melt track cross-sections produced via the National Institute for Standards and Technology (NIST) Additive Manufacturing Metrology Testbed (AMMT) [54]; and single-layer melting experiments performed on a commercial LPBF machine (EOS M290).

For the bare plate experiments, we utilised SS316 substrates (0.25" thick) that were sandblasted on the top for improved laser absorptivity. Laser scan lines had a 1 mm spacing and were printed perpendicular to the gas flow direction. The laser spot size was held constant at $4\sigma = 109\ \mu\text{m}$, and the scan speed at 1 m/s, with power settings of 60 W, 130 W, and 200 W, as published in [53].

The second set of training images derived from bare plate experiments was provided by NIST and is available publicly [30,31,54]. The experiments employed nickel superalloy 625 (IN625) substrates, and a wide range of laser process parameters [30].

Third, single powder layer melting experiments were performed using a commercial EOS M290 LPBF machine under standard operating conditions. High-precision etched SS316 single layer templates were used a high level of control over powder layer consistency. 15 – 45 μm SS316L powder was manually spread with a machinist blade of 1/8 in thickness into templates with depths ranging from 60 μm to 150 μm . The powder size distribution is $D_{10} = 22.6\ \mu\text{m}$, $D_{50} = 36.1\ \mu\text{m}$, $D_{90} = 56.0\ \mu\text{m}$ (Carpenter Technology). For more detail on the experimental method and single layer template preparation, the reader is referred to [55].

After laser melting, the post-processing of the melt tracks is similar across all three sources of experimental data, as schematically shown in Fig. 1. The substrate plates were cut perpendicular to the melt track direction with a low impact cutting method such as a diamond blade saw, then ground, polished, and subsequently etched with Aqua Regia (1pt water, 1pt nitric acid, 1pt hydrochloric acid) to reveal the metallographic microstructure and solidified melt pool boundaries.

Fig. 2 shows exemplary images from our training data, with these examples being selected to highlight the diversity of the data, for

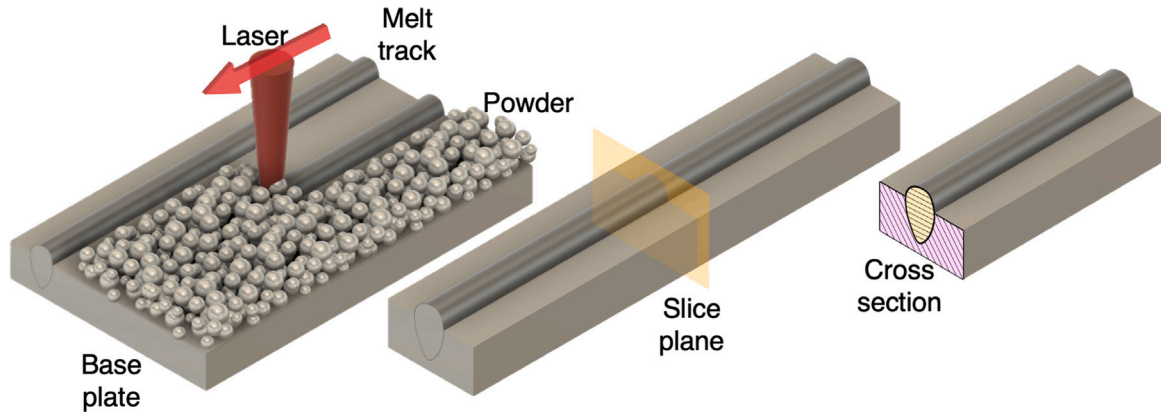


Fig. 1. Schematic of the experimental process flow from single layer melting to cross-sectioning. Many parallel single-tracks are typically melted in a single experiment, and subsequently cross-sectioned, and then imaged separately from the sample. The schematic shows sectioning of only single tracks.

instance: shallow vs. deep melt pools; a bare plate without new material being deposited vs. significant material being melted to the substrate; different levels of etching, colour, and contrast. Some of the images contain scale bars of different types and sizes, which the neural network must learn to ignore. From each of the cross-sections, geometrical features include the melt track depth, the amount of the melt track above the substrate surface line (i.e., new material being deposited) vs. below the substrate surface line, melt track width, cross-sectional area, wetting angle α , as well as wall angle β (see Fig. 3).

2.2. Generation of training data

The training data comprises of the raw images and the segmented (labelled) images. To speed up and standardise the manual segmentation of the large number of images needed for the training, we developed a software that uses morphological geodesic active contours (MGAC) [56] as the first step to generate masks for our images. The user sets an initial small diameter ellipse as a nucleus within the melt track to initiate the process. Based on an “energy field” defined by edge detection and subsequent blurring, the initial ellipse is iteratively ballooned, growing its size and conforming to the contour edge of the melt track from the inside out. The software outputs 7 different masks, each corresponding to a different set of hyperparameters. A user may then select the most visually accurate mask. Fig. 4 shows an example image and its corresponding masks generated by the active contours method.

For cases where none of these segmentation masks are satisfactory to the user to include in the training data set, we built a second software based on connected colour thresholding, which relies more on manual human intervention. The user can paint over a region in the image, for example within the melt track in this case, and the software selects all connected pixels with colour intensity values within a user-defined threshold of the selected pixels. Fig. 5 shows the different steps of using this tool with an example image. This is a more labour intensive method, but produces very accurate results because the melt pool boundaries have a colour that is distinct enough from the centre of the melt pool and from grain boundaries. Moreover, this is a one-time effort as these tools will not be required once we have a trained model. A simple flood fill takes care of all dislocations within the melt pool. Both these tools are available as an installable python package and software [52].

Using these tools, we generated 76 images, split into 62 training, 6 validation, and 8 testing images. Given the relatively small data set of images, we generate additional images to augment the data set using simple computer vision transformations. The parameters for these transformations are summarised in Table 1. Each augmented image is generated by randomly applying one of these transforms to an original

Table 1
Parameters used for image augmentation.

Operation	Range
Rotation	$\pm 20^\circ$
Width shift	5%
Height shift	5%
Shear	0.05 radians
Zoom	95–105%
Brightness	$\gamma = [0.2, 1.8]$
Vertical flip	50%
Horizontal flip	50%

image. We generate 15 augmented images for each image from the initial dataset. The augmented data in this 992 image dataset (62 real images and 62×15 augmented images) provides greater diversity to the training data, making it more robust to changes in size, contrast, resolution, and melt track location.

2.3. U-Net neural network

U-Net is an encoder–decoder architecture that has proven its versatility at image segmentation tasks [59]. Fig. 6 shows the specific architecture used. The left half is the encoder half, with multiple blocks each comprising of two convolutional layers with a 3×3 tile size and one 2×2 max pooling layer. Starting with 64 channels in the first block, we double the number of channels in each block, ending with 1024 channels after 4 blocks. At this stage, the input image has been reduced to a latent space representation, where only the pixels with features important to the segmentation task have been activated.

The right half of the convolutional neural network (CNN) decodes the latent space representation back to a 2D image. Here too there are 4 blocks, each halving the number of channels and comprising of one 2×2 upconversion layer followed by two convolutional layers with a 3×3 tile size. The upconversion layers also include a step where the output from the corresponding convolutional layer in the encoding half is copied over to reintroduce spacial information. The output from these layers passes through a 1×1 convolutional layer with a sigmoidal activation function. All the layers except the last one use a rectified linear unit (ReLU) activation function. We used binary cross entropy to calculate the loss, with the Root Mean Squared Propagation (RMSprop) optimiser to calculate the gradients:

$$L(x, z) = -(z \log(g(x)) + (1 - z) \log(1 - g(x))) \quad (1)$$

where x are the logits, z are the labels, and g is the sigmoid function. To find the optimal hyperparameters, we trained the model with batch sizes of 8, 16 and 32, and learning rates of 10^{-5} , 10^{-4} and 10^{-3} .

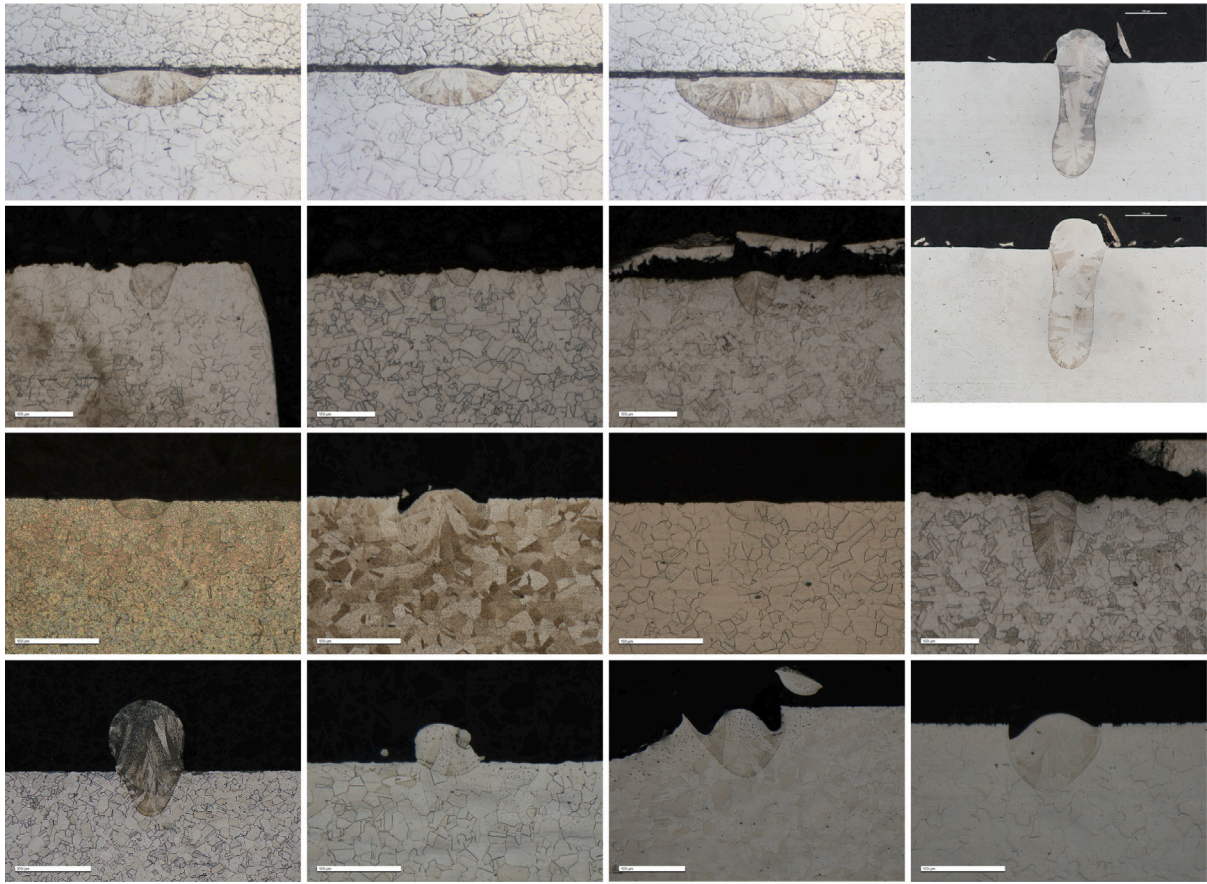


Fig. 2. Example micrographs of melt track cross-sections showing a diversity in the contrast, material, colour profile, and aspect ratio. The images present artefacts such as scale bars of different types and in various locations, excess powder, and an additional plate.

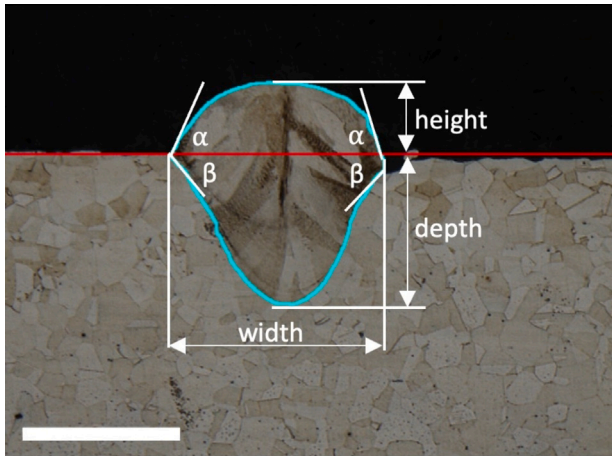


Fig. 3. Ideal melt track cross-section showing the extracted geometrical metrics: melt track depth, height, width, wetting angle α , wall angle β . The red line shows the baseline for the substrate surface and cyan line shows the border of the melt pool. The scale bar is 100 μm . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.4. Post-processing

From the binary segmented image, we extract the true surface and baseline first. If the intensity of the top half of the image is too high, we change it to have zero intensity; this tackles the case where there is

another sample in the top half of the image. We then identify the top edge using a Gaussian blur followed by a sobel filter. This is the true surface of the metal. To identify the baseline (refer to Fig. 3), we fit a line to the points in the vicinity of the melt pool.

Once we have the baseline, we identify the bounding box of the melt track. The height and depth of the melt track are calculated with respect to the baseline, and the width is the same as the bounding box.

To calculate the wetting and wall angles, we identify the point where the true surface intersects the melt track as the tangent point. There is one such point on each side of the melt track. We then identify the tangents to the melt pool at those points in the top and bottom directions, and calculate their angle with the baseline. We report the mean of the wetting and wall angles on the left and right sides.

3. Results and discussion

3.1. Hyperparameter optimisation

To find the optimal hyperparameters for training the model, we varied the learning rate and batch size, and evaluated the model using the pixel-wise accuracy and F1 score as metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{F1 Score} = \frac{2TP}{2TP + FP + FN} \quad (3)$$

where TP , TN , FP , and FN stand for true positives, true negatives, false positives and false negatives. In our study, we label the melt pool pixels as 1, i.e. positive, and the substrate and background pixels as 0, i.e. negative. The accuracy is just the proportion of pixels correctly classified. The F1 score, on the other hand, provides a measure of

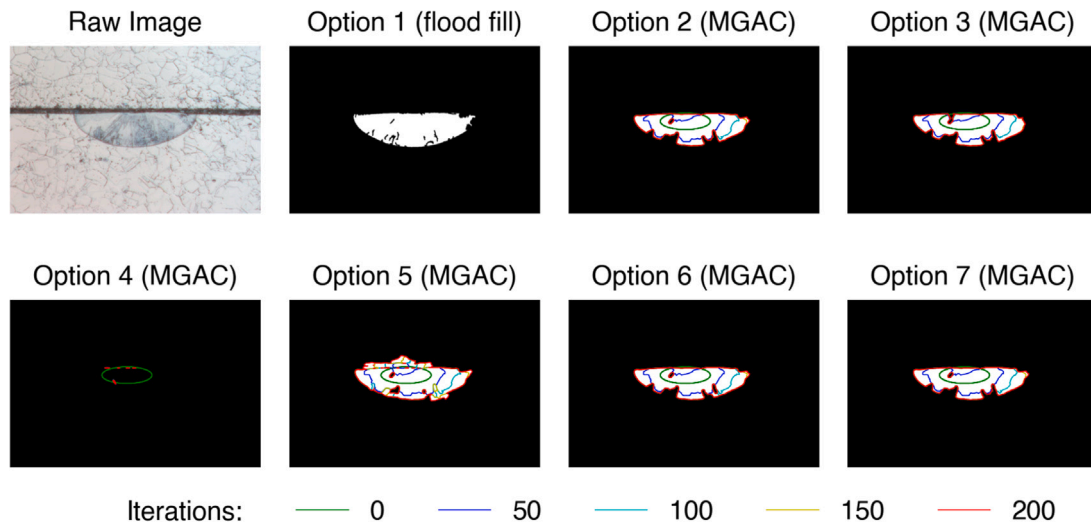


Fig. 4. Output from the software that uses morphological geodesic active contours (MGAC) to identify the melt pool boundary. The first figure is the raw image. The second figure is a flood fill of the central region after edge detection. Each of the remaining output figures correspond to a different set of hyperparameters for the MGAC based technique. Within each MGAC Option, a small number of the ballooning iterations is depicted using the colours in the legend at the bottom of the figure.

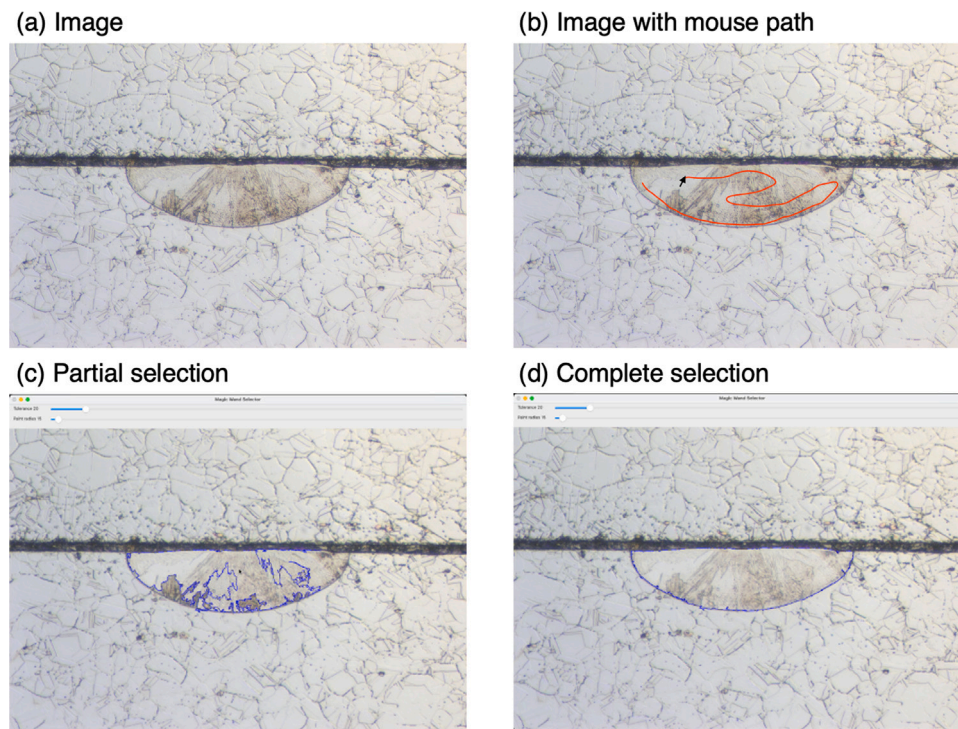


Fig. 5. Steps while using connected colour thresholding [57] to segment an image. (a) The raw image; (b) a schematic of the mouse path taken to select the whole region; (c) a snapshot of the region selected halfway through the mouse path in (b); (d) the complete selection.

how closely the prediction aligns with the ground truth. It achieves this by calculating the harmonic mean of two key metrics: recall and precision. Recall represents the fraction of actual melt pool pixels that the model correctly identifies, while precision indicates the fraction of pixels classified as melt pool that are indeed correct. Fig. 7 shows the mean accuracy and mean F1 scores achieved during training. These were obtained using batch sizes of 8, 16, and 32, and initial learning

rates of 10^{-5} , 10^{-4} and 10^{-3} . The best results were produced with a batch size of 16 and an initial learning rate of 10^{-4} .

3.2. Model performance

The trained model performs well on most cross-sectional images of LPBF melt tracks. It is capable of accurately identifying melt pools

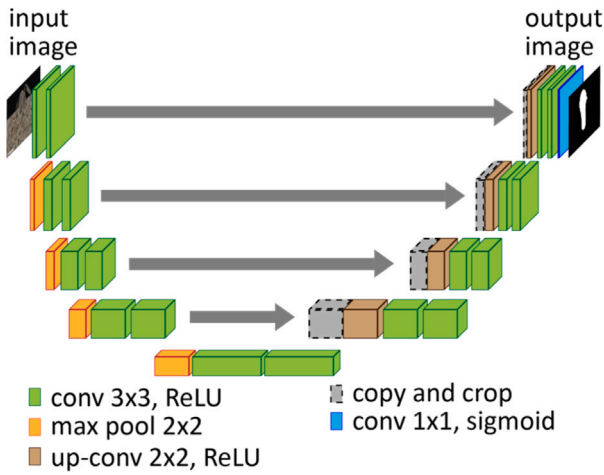


Fig. 6. The U-Net architecture used. The left half is the encoding half that reduces the input to a latent space representation where the relevant pixels are activated. The right half is the decoding half which brings the latent space representation back to 2D space with the copy and crop steps imparting contextual information.

Source: Adapted from [58].

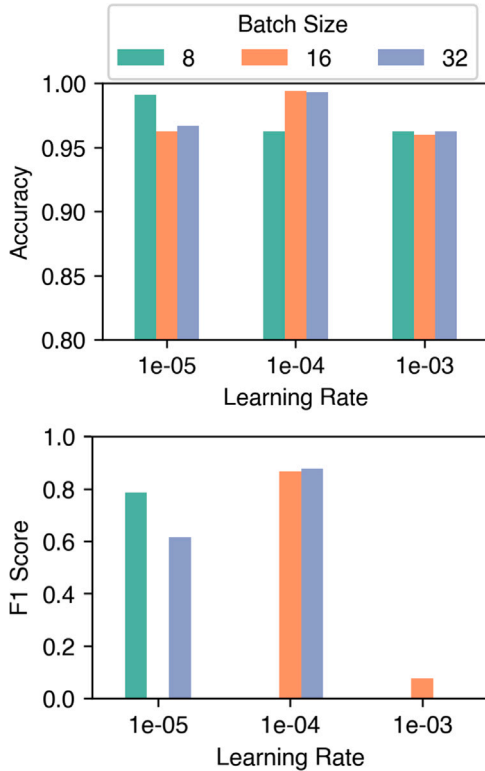


Fig. 7. Mean accuracy and F1 score for models trained with different combinations of learning rates and batch sizes all measured on the validation data set. A learning rate of 10^{-4} and a batch size of 16 gave the best results. Missing bars indicate zeros, i.e. the loss function could not be minimised.

in images having different contrasts and colour profiles, images of different materials, and images of tracks made on different machines. As shown in Fig. 8, both the training and validation losses decrease monotonically with training, and the difference between them is small, indicating that the model does not suffer from overfitting. The trained model has an accuracy of 0.995, an F1 score of 0.904, and an intersection over union (IoU) of 0.868 (all metrics are measured over the

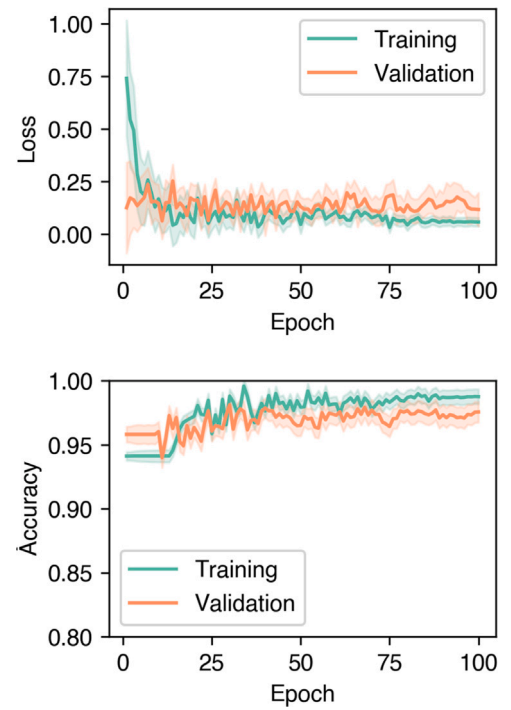


Fig. 8. Training and validation loss and accuracy vs. epoch. The values have been averaged over 3 initialisations on different train and validation splits. The bands represent ± 1 standard deviation.

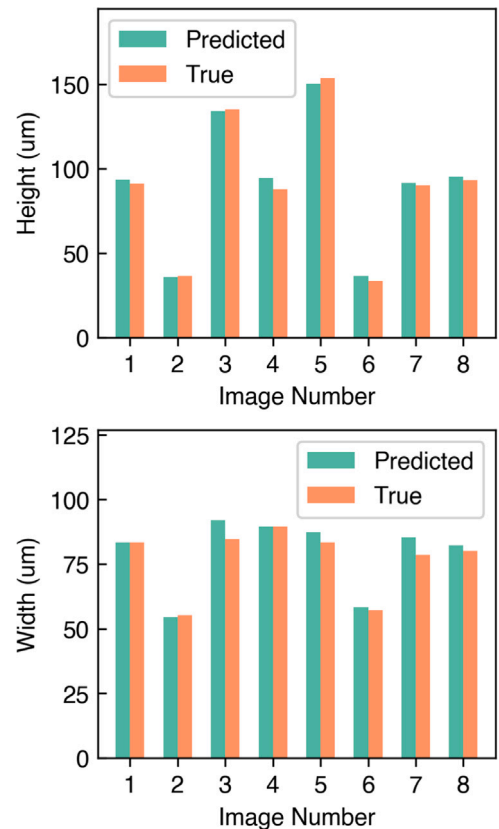


Fig. 9. True and predicted values for the height of the entire melt pool for 8 new images. The predicted values agree well with the true values.

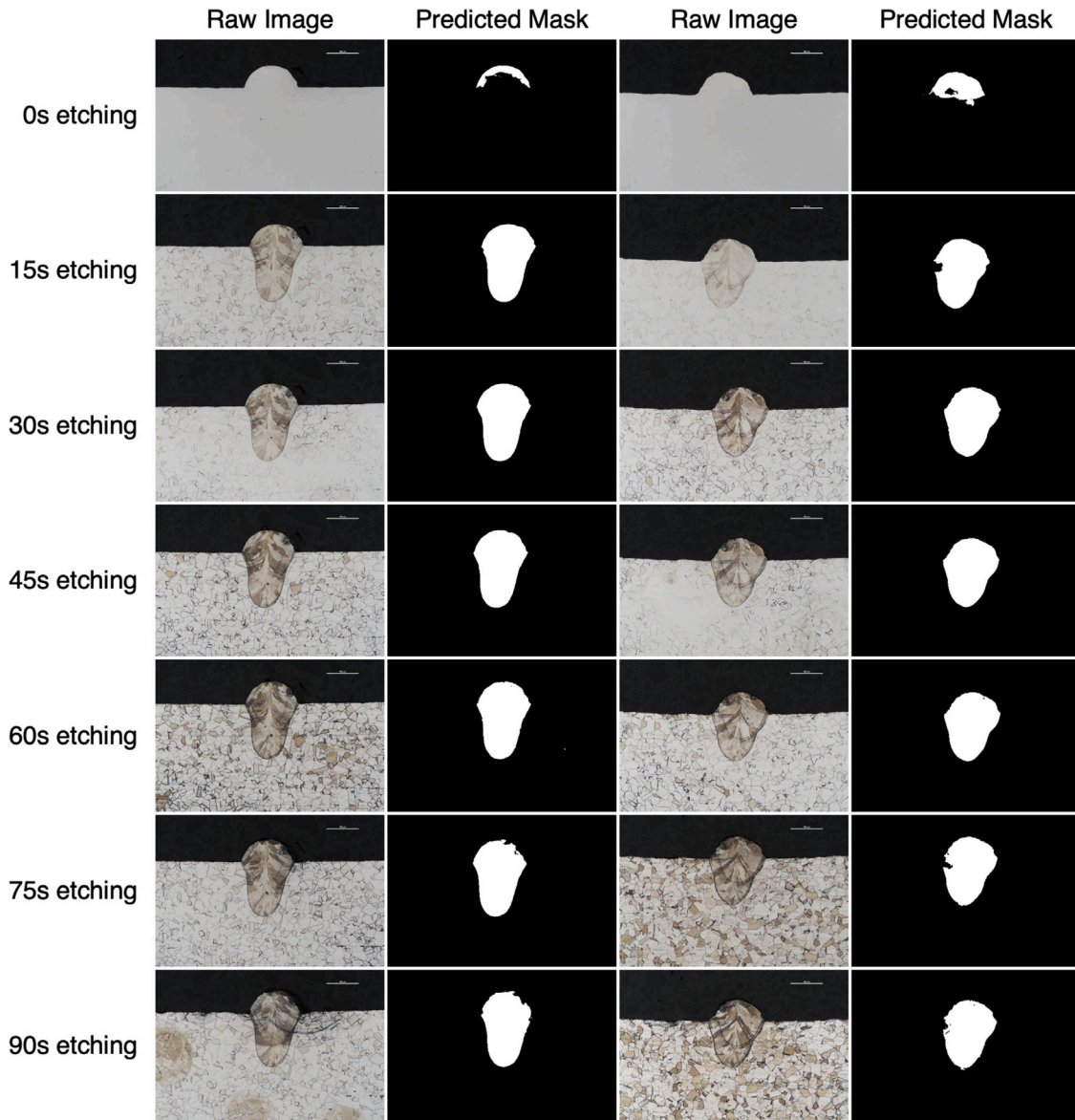


Fig. 10. Two samples imaged incremental etching with total time progressing from 0 s to 90 s. The first and second columns show the raw image and predicted mask for the first sample. The third and fourth columns show the same for the second sample. As is visible here, etching is necessary to reveal the melt track boundary, but the model is robust to various contrasts and colour profiles caused by different amounts of etching.

test data). IoU is a common metric used to assess cases where a single domain must be identified and is defined as

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (4)$$

Importantly, such high values could be achieved with relatively small training data due to the diversity and high quality of data used [58].

We show the performance of the model on 8 unseen images in Fig. 9. Overall, the predictions are in good agreement with the true values for both height and width. The mean absolute error (in percentage) for the height predictions is 2.84% and for the width predictions is 3.59%. These values are lower than the typical human error in image-based measurements [60–62] indicating good model performance.

Fig. 10 shows the performance of the model on two samples, both etched from 0 s (no etching) to 90 s (45 s is the normal etching time). Here we observe that the model performs extremely well from 15 s onwards. This shows that the model is capable of segmenting a wide range of contrasts introduced by different levels of etching and different experimental protocols. We expect the model to be used for future high throughput analysis of LPBF melt pools. In follow-up work, we

have applied the model as trained here to automatically extract melt pool dimensions from a dataset of cross-sections at wide-ranging LPBF parameters, and produce scaling relationships between dimensionless parameters [55].

However, the model has some limitations. It occasionally fails to accurately segment the images in the following cases: (1) the image is extremely out of focus; (2) the boundary of the melt pool is visibly hard to discern; (3) the image contains grain boundaries that are similar in contrast and texture to the edge of the melt pool; (4) the image's colour profile markedly differs from the training data (refer to Fig. S1 for exemplary images). The first 3 limitations stem from errors in the experimental process. (1) is a direct result of poor focus during imaging — this leads to a low signal-to-noise ratio which cannot be fixed computationally. (2) and (3) are a result of under-etching and over-etching the sample, respectively. Both these can be addressed by tuning the etch duration. (4) is usually a characteristic of using a microscope that provides a completely different colour profile compared to any of the images in the training data. This problem can be addressed by

adding some images from this new source to the training dataset and retrain the model.

3.3. Potential future directions

First, the model can be expanded to more rigorously consider defects within melt tracks, such as subsurface pores. Keyhole pores, which are typically circular in cross-section, are some of the most common artefacts in the images. The current model handles keyhole pores differently based on their location. If a pore is entirely within the melt pool, the pore is disregarded and considered as part of the melt pool. However, if the pore lies along the melt pool boundary, it is considered outside the pool. Another printing regime involves ‘balling’, which is challenging to detect from cross-sectional images. Some telltale signs of balling include wetting angles exceeding 90° and most of the melt pool height being above the surface. The model is capable of accurately segmenting those images as well. In future work, the model could be expanded to specifically identify and label these defects and calculate metrics such as pore size and position.

A future direction for the development of this model is its use for multi-track images, namely raster scans for a single layer of LPBF, which have parallel melt tracks with partial overlap. The model is able to accurately segment some images with multiple melt pools, despite being trained on no such images (refer to Fig. S2 for some exemplary images). Addition multi-track images to the training data set would likely improve the accuracy of the model’s prediction on such images.

The model could potentially also be made more robust by exploring new image augmentation techniques such as changes to the hue and saturation, and adding more augmented data. It is possible a model trained with such a dataset is robust to more colour profiles stemming from new microscopy equipment.

Given the challenges of generating training data and training a neural network on one method, it also is of interest to apply the current trained model to a different AM process. We tested the model’s performance on a few cross-sectional images of samples made by Directed Energy Deposition (DED). The model is able to accurately segment some but not all images (refer to Fig. S2 for some exemplary images). This indicates the prospect of transfer learning [63], where our model is used as the starting point for a model geared to DED and trained with a much smaller data set. Separately, a new training set, and trained model could be created for DED or for other AM processes with characteristic image features and dimensions of interest.

4. Conclusion

In this study, we demonstrated the use of a U-Net-based CNN to effectively segment micrographs of the cross-sections of single-track LPBF melt pools. The trained model performs with an accuracy greater than 99% and an F1 score greater than 90% despite the relatively small size of the training data set. We show the model’s robustness to artefacts, changes in contrast from etching, and changes in colour profile and resolution from different microscopes. We believe the ability of a single model to effectively handle this heterogeneous data is important for practical adoption in research and production scenarios. We also demonstrate the use of post-processing techniques to collect valuable quantitative metrics from the predicted masks, such as width, height, depth, and wetting and wall angles. We expect that this automated segmentation technique in conjunction with the printing parameters will help expedite the optimisation of high-quality and reliable LPBF. We also highlight some future directions to improve the model and the scope of transfer learning.

CRedit authorship contribution statement

Aagam Shah: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Reimar Weissbach:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **David A. Griggs:** Data curation. **A. John Hart:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition. **Elif Ertekin:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition. **Sameh Tawfik:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sameh Tawfik reports financial support was provided by National Science Foundation. Aagam Shah reports financial support was provided by National Science Foundation. Elif Ertekin reports equipment, drugs, or supplies was provided by Pittsburgh Supercomputing Center. Elif Ertekin reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge support from the National Science Foundation (NSF), USA through award #1720701 and #1922758. This work used Bridges-2 at the Pittsburgh Supercomputing Center through allocation MAT220011 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by National Science Foundation, USA grants #2138259, #2138286, #2138307, #2137603, and #2138296. The authors also thank Dr. Brandon Lane, Dr. Lyle Levine, and Dr. Jordan Weaver from the National Institute of Standards and Technology, USA for images used to train the model, along with their guidance.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmapro.2025.09.043>.

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